STAT 2020 Discussion Slides

Week 8: Exponential, Poisson, and Z-table

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Index (Outline)

1. Week 8 Goals
2. Exponential vs Poisson
3. Memoryless Property
4. Continuous RV Review
5. Using the Z-table
6. Practice (Quick Drills)

Agenda

- Exponential vs. Poisson: what is modeled and key formulas
- Identify λ correctly in exponential models
- Memoryless property practice
- Continuous random variable review: f, F, $\mathbb{E}(X)$, $\mathbb{V}(X)$
- How to use the Z-table for normal probabilities
- Attendance check

Basic Formulas (Must Know)

Poisson (count in a window):

$$X \sim \text{Pois}(\mu), \qquad P(X = k) = \frac{e^{-\mu}\mu^k}{k!}, \quad k = 0, 1, 2, \ldots$$

$$\mathbb{E}(X) = \mu, \qquad \mathbb{V}(X) = \mu.$$

Exponential (waiting time):

$$X \sim \text{Exp}(\lambda), \qquad f(x) = \lambda e^{-\lambda x}, \ x > 0$$

$$F(x) = 1 - e^{-\lambda x}, \quad x > 0, \qquad \mathbb{E}(X) = \frac{1}{\lambda}, \quad \mathbb{V}(X) = \frac{1}{\lambda^2}.$$

Exponential vs. Poisson: Quick Comparison

Distribution	Random Variable	Typical Question
Poisson	count	"How many calls in 10 minutes?"
Exponential	waiting time	"How long until next call?"

If arrivals follow a Poisson process with rate λ:

$$N_t \sim \mathrm{Pois}(\lambda t), \qquad T \sim \mathrm{Exp}(\lambda).$$

How to Find λ in Exponential Questions

If mean waiting time is m, then

$$\mathbb{E}(X) = \frac{1}{\lambda} = m \quad \Longrightarrow \quad \lambda = \frac{1}{m}.$$

Example: mean wait $= 8$ minutes

$$\lambda = \frac{1}{8} \text{ per minute}, \qquad f(x) = \frac{1}{8} e^{-x/8}.$$

Memoryless Property of Exponential

For $X \sim \mathrm{Exp}(\lambda)$, for any $s, t > 0$:

$$P(X > s + t \mid X > s) = P(X > t).$$

Equivalent survival form:

$$P(X > u) = e^{-\lambda u}, \quad u > 0.$$

So

$$P(X > s + t \mid X > s) = \frac{e^{-\lambda(s+t)}}{e^{-\lambda s}} = e^{-\lambda t}.$$

Memoryless Example

Suppose $X \sim \mathrm{Exp}(\lambda)$ with mean 6 minutes. Then $\lambda = \frac{1}{6}$.

$$P(X > 10 \mid X > 4) = P(X > 6) = e^{-(1/6)\cdot 6} = e^{-1} \approx 0.3679.$$

Key point: once the student has already waited 4 minutes, only the extra 6 minutes matters.

Continuous RV Basics

$$F_X(x) = P(X \leq x) = \int_{-\infty}^{x} f_X(t)\,dt, \qquad f_X(x) = F_X'(x) \text{ (where differentiable).}$$

$$\mathbb{E}(X) = \int_{-\infty}^{\infty} x f_X(x)\,dx, \qquad \mathbb{E}(X^2) = \int_{-\infty}^{\infty} x^2 f_X(x)\,dx.$$

$$\mathbb{V}(X) = \mathbb{E}(X^2) - [\mathbb{E}(X)]^2.$$

Practice CDF

$$F(x) = \begin{cases} 0, & x < 0, \\ \dfrac{e^x - 1}{e - 1}, & 0 \le x < 1, \\ 1, & x \ge 1. \end{cases}$$

Differentiate by interval:

$$f(x) = \begin{cases} \dfrac{e^x}{e - 1}, & 0 < x < 1, \\ 0, & \text{otherwise.} \end{cases}$$

This is an increasing e^x-shaped curve (not triangular).

Mean and Variance (Quick Result)

For the previous density:

$$\mathbb{E}(X) = \frac{1}{e-1} \approx 0.582, \qquad \mathbb{V}(X) \approx 0.0796.$$

Also,

$$\mathbb{E}(X^2) = \frac{e-2}{e-1}, \qquad \mathbb{V}(X) = \mathbb{E}(X^2) - [\mathbb{E}(X)]^2.$$

Standard Normal Setup

If $X \sim \mathrm{N}(\mu, \sigma^2)$, define

$$Z = \frac{X - \mu}{\sigma} \sim \mathrm{N}(0, 1).$$

Z-table usually gives:

$$\Phi(z) = P(Z \leq z).$$

Convert first, then read table.

Z-table Example 1

Let $X \sim \mathrm{N}(70, 10^2)$. Find $P(X \leq 85)$.

$$z = \frac{85 - 70}{10} = 1.5.$$

$$P(X \leq 85) = P(Z \leq 1.5) = \Phi(1.5) \approx 0.9332.$$

Z-table Example 2

Let $X \sim \mathrm{N}(70, 10^2)$. Find $P(60 < X < 80)$.

$$P(60 < X < 80) = P(-1 < Z < 1) = \Phi(1) - \Phi(-1).$$

Using the table:

$$\Phi(1) \approx 0.8413, \quad \Phi(-1) \approx 0.1587$$

$$P(60 < X < 80) \approx 0.6826.$$

Practice 1: Identify Distribution

- Number of emails received in 1 hour: Poisson or Exponential?
- Time until next email arrives: Poisson or Exponential?

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Answer:

- Count in a fixed window $\rightarrow$ Poisson
- Waiting time $\rightarrow$ Exponential

Practice 2: Find λ

Waiting time X is exponential with mean 10 minutes.

- Find λ
- Write $f(x)$

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Waiting time X is exponential with mean 10 minutes.

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- Write $f(x)$

Answer:

$$\lambda = \frac{1}{10}, \qquad f(x) = \frac{1}{10}e^{-x/10},\ x > 0.$$

Practice 3: Memoryless

$X \sim \text{Exp}(1/8)$. Compute $P(X > 12 \mid X > 5)$.

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$X \sim \text{Exp}(1/8)$. Compute $P(X > 12 \mid X > 5)$.

$$P(X > 12 \mid X > 5) = P(X > 7) = e^{-7/8} \approx 0.4169.$$

Practice 4: CDF to PDF

If

$$F(x) = \begin{cases} 0, & x < 0, \\ x^2, & 0 \leq x < 1, \\ 1, & x \geq 1, \end{cases}$$

find $f(x)$.

Practice 4: CDF to PDF

If

$$F(x) = \begin{cases} 0, & x < 0, \\ x^2, & 0 \leq x < 1, \\ 1, & x \geq 1, \end{cases}$$

find $f(x)$.

$$f(x) = \begin{cases} 2x, & 0 < x < 1, \\ 0, & \text{otherwise.} \end{cases}$$

Practice 5: Z-table

If $Z \sim \mathrm{N}(0,1)$, compute $P(Z \leq 0.84)$.

Practice 5: Z-table

If $Z \sim \mathrm{N}(0,1)$, compute $P(Z \leq 0.84)$. Use table:

$$P(Z \leq 0.84) = \Phi(0.84) \approx 0.7995.$$

Class Reminder: Exam 1 Questions

- We can spend a few minutes on Exam 1 questions in this class.
- Focus on common mistakes and core concepts.
- Keep discussion short, then return to this week's topics.

Take-Home Messages

- Poisson models counts; Exponential models waiting times.
- For Exponential, identify λ from the mean: $\lambda = 1/\mathbb{E}(X)$.
- Use memoryless property to simplify conditional waiting-time problems.
- For normal probability, standardize first, then use the Z-table.

In-Class Attendance Check (Quick)

Waiting time X (minutes) follows $\mathrm{Exp}(\lambda)$ with mean 10.
A student has already waited 3 minutes. What is

$$P(X > 8 \mid X > 3)?$$

Please submit: Full Name; Section Number; Your Answer

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Answer:

$$P(X > 8 \mid X > 3) = P(X > 5) = e^{-0.5} \approx 0.6065.$$